

Chapter 1: Picturing Distributions with Graphs

School of Mathematical and Computational Sciences
University of Prince Edward Island

Ali Raisolsadat

Part I: Exploring Data

The Course Roadmap

I: Exploring Data

- Ch 1–2: Distributions & Numerical Summaries
- Ch 3: Scatterplots & Correlation
- Ch 4: Regression

II: Probability & Distributions

- Ch 9–10: Probability Rules
- Ch 11: Normal Distributions
- Ch 12: Binomial / Poisson

III: Foundations of Inference

- Ch 6: Samples & Observational Studies
- Ch 13: Sampling Distributions
- Ch 14: Confidence Intervals & p -values
- Ch 15: Inference in Practice

IV: Specific Inference Procedures

- Ch 17: Inference about a Population Mean
- Ch 19: Inference about a Population Proportion

What Is Statistics?

Definition

Statistics is the science of learning from data.

Ernest Rutherford

All science is either physics or stamp collecting.

Statistics gives us tools to:

- collect information;
- organize and display data;
- summarize patterns, compare groups and use data to answer questions.

Statistics Is More Than Formulas

A statistical thinker asks: *What does the data show, what does it not show and how confident should we be in the conclusion?*

Statistics Is Everywhere

- **Biology:** Do plants given fertilizer grow taller than plants that do not receive it?
- **Chemistry:** Does changing the catalyst increase reaction yield?
- **Psychology:** Do students who sleep longer tend to report better concentration?
- **Economics:** Are grocery prices increasing over time, or was it only COVID that distort our impression?
- **Artificial Intelligence:** Does the training dataset fairly represent the population where the AI system will be used?

Why Statistics Still Matters

LLMs and AI Can Help Us

- organize data quickly;
- calculate summaries;
- generate visualizations;
- identify possible patterns.

But We Still Must Ask

- Is this the right graph?
- Is the scale fair?
- Are there errors or unusual values?
- Does the conclusion actually follow from the data?

Anatomy of a Dataset

The Rectangular Data Idea

In a typical dataset **rows represent individuals or cases** and **columns represent variables**.

Plant ID	Fertilizer	Germinated?	Leaves	Height (cm)
P01	None	Yes	5	12.4
P02	None	Yes	6	13.1
P03	Fertilizer A	Yes	8	16.7
P04	Fertilizer A	No	0	0.0
P05	Fertilizer B	Yes	7	15.9

First Question for Any Dataset

What does one row represent?

Individuals, Variables and Identifiers

From the plant dataset:

- **Individuals:** the individual plants.
- **Variables:**
 - Fertilizer Group
 - Germinated?
 - Number of Leaves
 - Height
- **Plant ID:** mainly an identifier used to keep track of each plant.

Definition

A **variable** is a characteristic recorded for each individual that can take different values.

Two Fundamental Types of Variables

Qualitative (Categorical)

Places an individual into a group or category.

- Blood type: A, B, AB, O
- Fertilizer group: None, A, B
- Reaction outcome: Complete, Partial, None
- University program

Quantitative (Numerical)

Records a numerical amount for which arithmetic is meaningful.

- Plant height (cm)
- Reaction time (ms)
- Temperature ($^{\circ}\text{C}$)
- Mass (g)
- Price (\$)

Categorical Variables: Nominal vs. Ordinal

Nominal

Categories have **no natural order**.

- Blood type
- Treatment group
- Reaction outcome type
- Academic program

Ordinal

Categories have a **meaningful order**.

- Low / Medium / High
- Poor / Fair / Good / Excellent
- Mild / Moderate / Severe

The labels describe groups; the labels themselves are not measurements.

Quantitative Variables: Discrete vs. Continuous

Discrete: Usually Counting

- Number of leaves
- Number of bacterial colonies
- Number of customers
- Number of incorrect answers

Continuous: Usually Measuring

- Plant height
- Body mass
- Reaction time
- Temperature
- Concentration

Count or Measure?

Question	Usually Produces	Example
How many?	Discrete quantitative	Number of leaves
How much?	Continuous quantitative	Plant mass
Which group?	Categorical	Fertilizer type
What level/rating?	Ordinal categorical	Low / Medium / High

Good Starting Point

This rule is not perfect for every advanced application but it works well for most introductory examples.

From Variable Type to Graph Type

Situation	Good Starting Graph	Main Question
One categorical variable	Bar graph	Which categories are most or least common?
Parts of one whole	Pie chart	How is one whole divided?
Small quantitative sample	Dotplot	What are the individual values and overall pattern?
Larger quantitative sample	Histogram	What is the shape of the distribution?
Measurements over time	Time plot	Is there a trend, cycle, or sudden change?

The best graph depends on both the variable type and the question.

The Distribution of a Categorical Variable

The **distribution** tells us what values a variable takes and how often they occur. For categorical data, we summarize with:

- **Frequency (count):** number of observations in each category.
- **Relative frequency (proportion):**

$$\text{Proportion} = \frac{\text{Count}}{n}.$$

- **Percentage:** proportion $\times 100\%$.

Example

If 18 of 40 students choose biology as their preferred science elective,

$$\frac{18}{40} = 0.45 = 45\%.$$

Example 1.1: Reaction Outcomes in a Chemistry Lab

Suppose 50 reaction trials are classified as follows:

Reaction Outcome	Count	Percentage
Complete reaction	28	56%
Partial reaction	15	30%
No visible reaction	7	14%
Total	50	100%

Table or Graph?

A table gives exact values. A graph makes the overall pattern easier to see quickly.

Bar Graphs: The Default Choice for Categorical Data

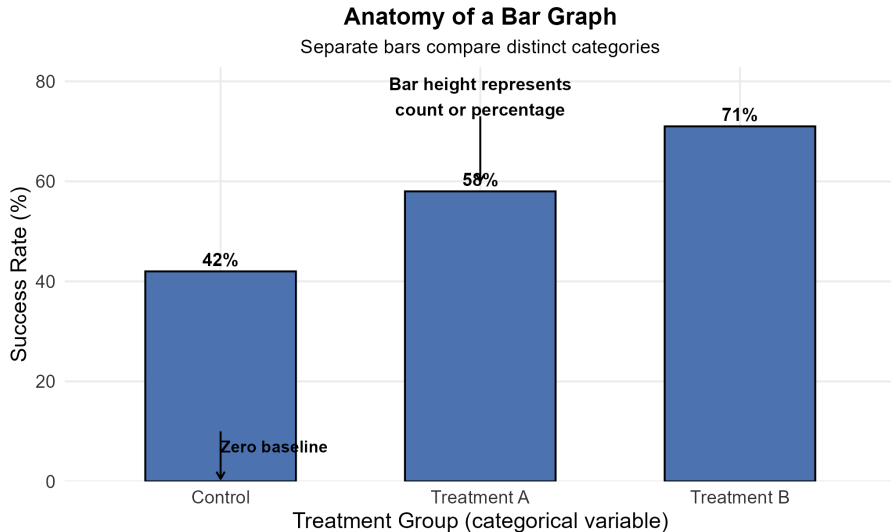
A **bar graph** represents each category with a separate bar.

- Bar height can show **count, proportion, or percentage**.
- Bars are **separated** because the categories are distinct.
- Nominal categories can often be reordered to make comparisons easier.
- Ordinal categories should usually keep their natural order.

When Reading a Bar Graph

Which category is largest? Smallest? Are any similar? Are the values counts or percentages?
Does the vertical axis start at zero?

Anatomy of a Bar Graph



Pie Charts: Parts of One Whole

A **pie chart** divides a single whole into categorical slices. Use a pie chart only when:

1. all categories belong to the **same group**;
2. categories are **mutually exclusive**;
3. categories represent the **entire sample population**; and
4. there are only a few categories.

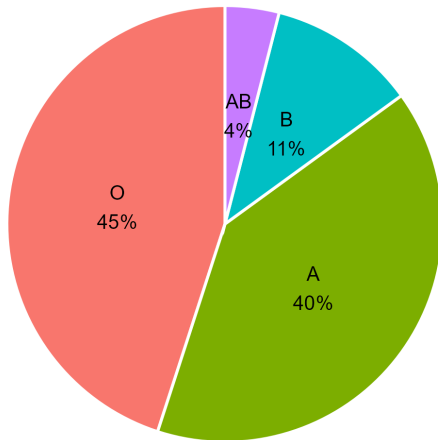
Practical Rule

A bar graph is usually the safer default because comparing lengths is easier than comparing angles and areas.

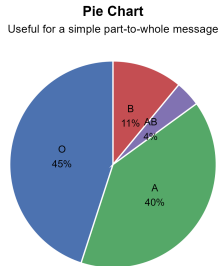
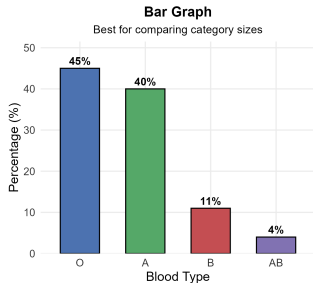
Anatomy of a Pie Chart

Anatomy of a Pie Chart

Every slice is part of the same 100% whole



Bar Graph vs. Pie Chart



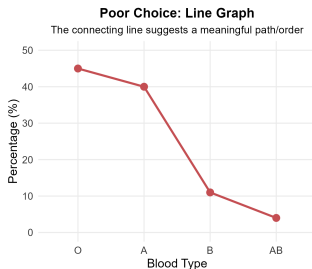
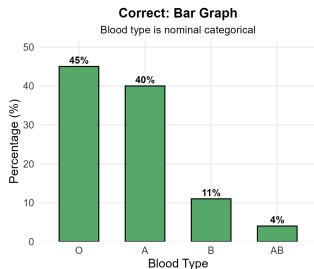
Both displays show the same blood-type percentages.

- The **pie chart** emphasizes part-to-whole structure.
- The **bar graph** makes small differences easier to compare.

Recommendation

Use a pie chart only when the part-to-whole relationship is the main message.

Using a Line Graph for Nominal Categories



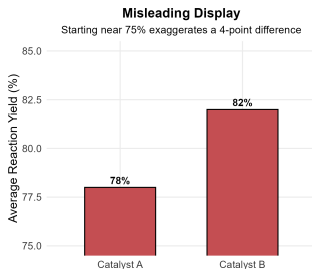
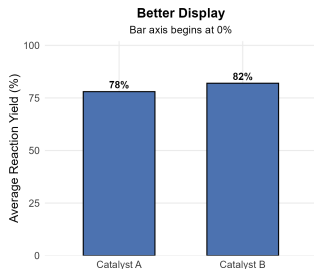
Blood type is a **nominal categorical variable**.

- There is no continuous path from O to A to B to AB.
- Connecting categories with lines suggests a relationship that does not exist.

Rule

Use lines when **order or continuity matters**, especially time.

Truncating the Axis of a Bar Graph



Catalyst A: **78% yield**

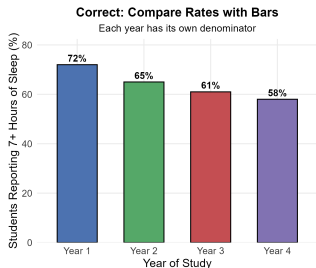
Catalyst B: **82% yield**

- Actual difference: **4 percentage points.**
- Starting the axis near 75% makes the bars appear dramatically different.

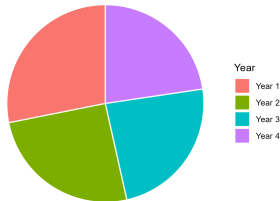
Bar-Graph Rule

Because bar length represents magnitude, the vertical axis should **usually begin at zero.**

A Pie Chart with Different Denominators



Wrong: A Pie Chart Creates a Fake Whole
72%, 65%, 61%, and 58% are separate group rates



Suppose the percentages sleeping at least 7 hours are:

- Year 1: 72%
- Year 2: 65%
- Year 3: 61%
- Year 4: 58%

Each percentage comes from a **different group**.

Correct Choice

Compare the group rates with a **bar graph**, not a pie chart.

A Quick Fair-Graph Checklist

Before trusting a graph, ask:

- Does the graph type match the **variable type**?
- Are the **axes** clearly labeled?
- Are the **units** shown?
- Does a bar graph begin at **zero**?
- Is anything important hidden by the **scale, bin width or ordering**?
- Does the title describe the data without exaggerating the conclusion?

Remember

A graph is an argument about the data. Design choices can make that argument more honest or more misleading.

From Numerical Measurements to a Distribution

Quantitative datasets may contain many distinct values:

17.8, 18.1, 18.3, 18.4, 18.7, ...

A separate bar for every exact value would quickly become difficult to read.

The Histogram Solution

Group nearby numerical values into intervals called **bins** or **classes** and count how many observations fall in each bin.

How a Histogram Works

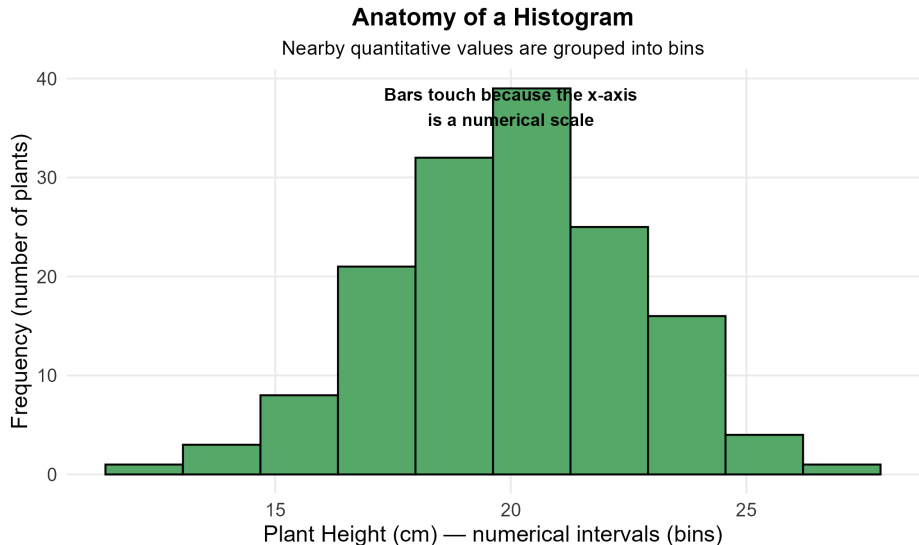
A histogram:

1. divides the numerical scale into intervals;
2. counts how many observations fall in each interval; and
3. draws bars whose heights show those frequencies.

Crucial Difference from a Bar Graph

Histogram bars are adjacent to each other because the horizontal axis is a continuous numerical scale.

Anatomy of a Histogram



Bar Graph vs. Histogram

Feature	Bar Graph	Histogram
Variable type	Categorical	Quantitative
Horizontal axis	Category labels	Numerical intervals
Bars	Separated	Adjacent
Order	May be rearranged	Must follow numerical order
Main purpose	Compare categories	Show distribution shape

Do not choose between them based only on how the graph looks. Start with the variable type.

Example 1.2: Body Lengths of 44 Great White Sharks

Marine biologists measured the body lengths of $n = 44$ great white sharks.

A Few of the Measurements (feet)

18.7, 12.3, 18.6, 16.4, 15.7, 18.3, 14.6, 15.8, 14.9, 17.6, ...

Rather than draw 44 separate bars, we group nearby values into length classes.

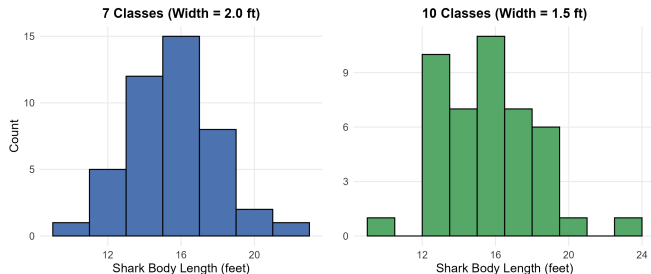
- Example class: 13.1 to 15.0 ft
- Number of sharks in that class: 12
- Approximate percentage: 27%

Example 1.2: Shark Length Frequency Table

Length Class (feet)	Count	Approx. %
9.1 to 11.0	1	2%
11.1 to 13.0	5	11%
13.1 to 15.0	12	27%
15.1 to 17.0	15	34%
17.1 to 19.0	8	18%
19.1 to 21.0	2	5%
21.1 to 23.0	1	2%
Total	44	99%*

*Percentages total 99% because of rounding.

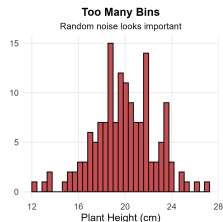
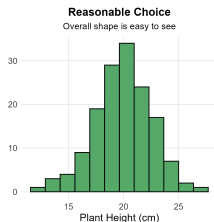
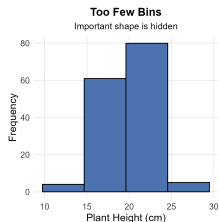
Example 1.2: The Same Shark Data, Different Bins



Both histograms use the same 44 shark lengths.

- The main cluster remains around the middle lengths.
- Changing bin width changes the amount of visible detail.
- The smallest and largest observations become more or less obvious.

The Bin Width Problem



- **Too few bins:** important features are smoothed away.
- **Too many bins:** random noise can look meaningful.
- **Reasonable width:** shows structure without manufacturing a story.

Good Practice

Inspect a few reasonable choices and check whether the main interpretation stays similar.

Break Time



How to Describe a Quantitative Distribution: SOCS

When asked to describe a histogram use **SOCS**:

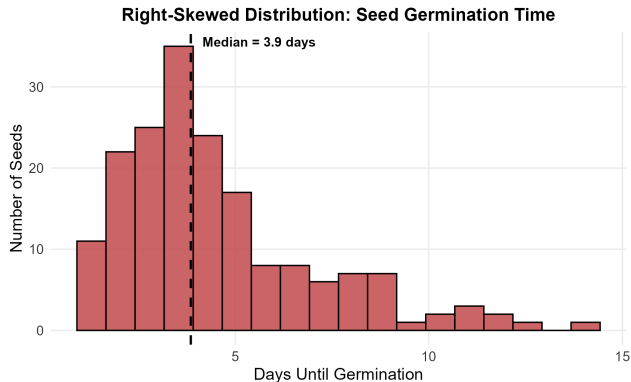
S – Shape: Is it symmetric, skewed left, skewed right, unimodal, or bimodal?

O – Outliers: Are any observations far from the overall pattern?

C – Center: Where is a typical or middle value?

S – Spread: How variable are the observations? What is the approximate range?

Shape: Symmetry, Skewness and Peaks



Common shapes include:

- **symmetric;**
- **skewed right;**
- **skewed left;**
- **unimodal;**
- **bimodal.**

Important

Skewness is named for the direction of the **long tail**, not the location of the tallest bar.

Outliers, Center and Spread

Outliers

Values far from the overall pattern. They may be genuine or may indicate an error.

Center

A typical or middle value. For now, estimate visually; Chapter 2 introduces mean and median.

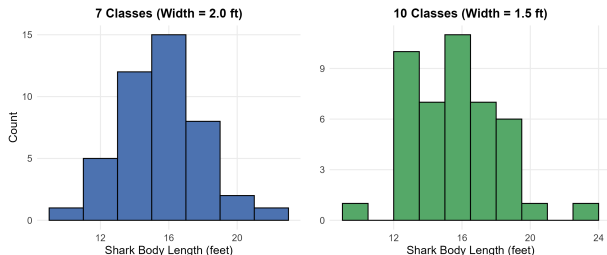
Spread

How much the values vary. A narrow distribution has less spread than a wide one.

Never Delete an Observation Just Because It Looks Unusual

Investigate the value, its units, and how it was recorded before deciding what to do with it.

Example 1.3: SOCS for Shark Lengths



- **Shape:** roughly unimodal and fairly symmetric through the center.
- **Outliers:** 9.4 ft and 22.8 ft appear separated from much of the data.
- **Center:** approximately 15–16 ft.
- **Spread:** 9.4 ft to 22.8 ft.

Goal

At this stage, learn to **read the distribution**; exact numerical summaries come next chapter.

Dotplots: Showing Individual Values

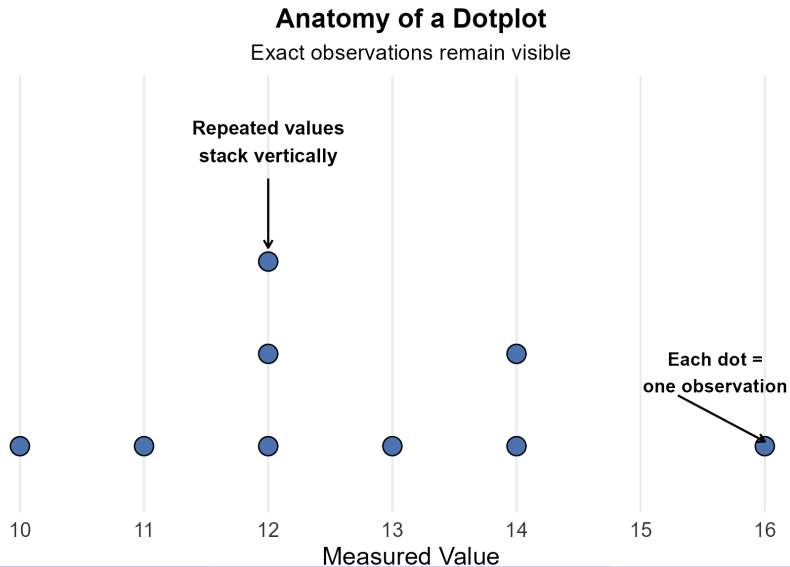
A histogram groups observations into bins. For a smaller dataset, we often want to see every value. A **dotplot** places one dot for each observation above a numerical scale.

- Individual values remain visible.
- Repeated values can be stacked.
- Clusters, gaps, center, spread, and outliers are easy to inspect.
- Small groups can be compared directly.

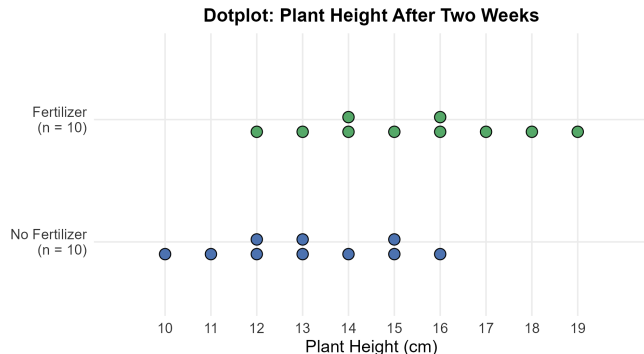
When?

The sample is small or moderate and the individual observations are still readable.

Anatomy of a Dotplot



Example 1.4: Plant Height With and Without Fertilizer



Each dot represents one plant.

- Fertilizer plants tend to be taller.
- The groups still overlap.
- No single point is extremely separated from the rest.

Careful Wording

The graph describes the **observed samples**. By itself, it does not prove what will happen for every future plant.

Dotplot vs. Histogram

Dotplot	Histogram
Shows individual observations	Groups observations into bins
Excellent for smaller samples	Excellent for larger samples
Repeated values remain visible	Overall shape is easier with dense data
Sample size can often be counted	Exact values are not visible

No Magic Cutoff

There is no universal rule that “30 observations means histogram.” Choose the display that communicates the data most clearly.

Time Plots: When Order Matters

A histogram or dotplot tells us **what values occurred**. It does not tell us **when they occurred**. When chronological order matters (such as time series), use a **time plot**:

- **x-axis:** time;
- **y-axis:** measured variable;
- points are commonly connected in chronological order.

Look for:

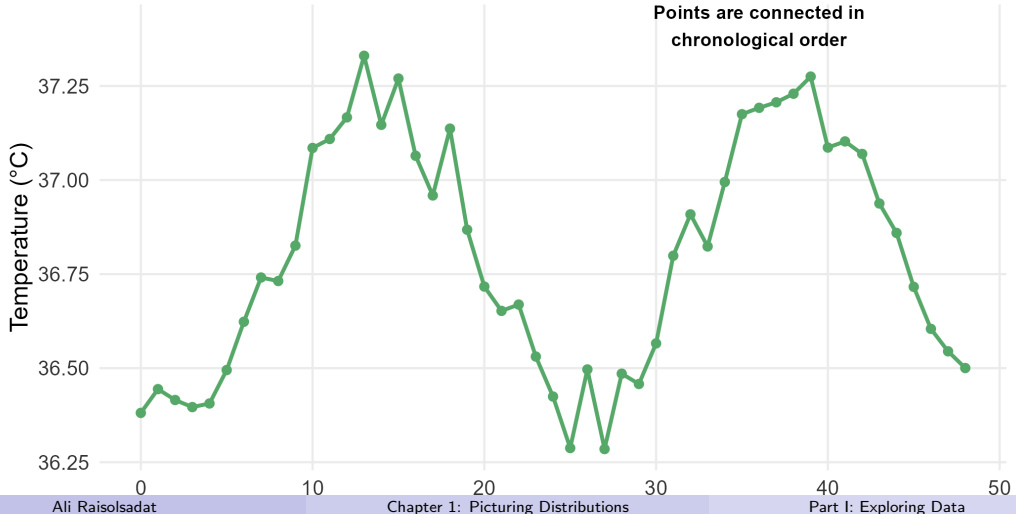
- **trend**
- **cycle**
- **sudden change**
- **unusual observation**

Anatomy of a Time Plot

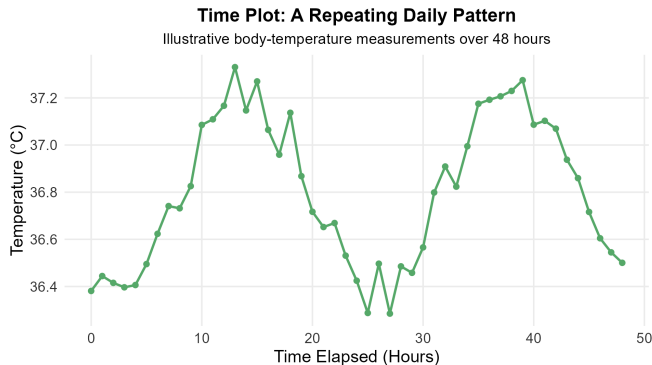
Anatomy of a Time Plot

Time always appears on the horizontal axis

Points are connected in
chronological order



Example 1.5: A Repeating Daily Pattern



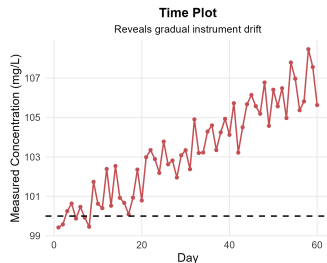
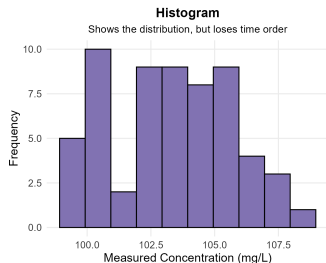
The measurements rise and fall in a repeating pattern over approximately 24 hours.

- A histogram could summarize the temperatures.
- But it would hide **when** the high and low values occurred.

Time-Plot Advantage

Time order makes cycles and trends visible.

Example 1.6: A Histogram Can Hide Instrument Drift



The two panels contain the **same measurements**.

- Histogram: what concentration values occurred?
- Time plot: when did they occur?
- The time plot reveals a gradual upward drift.

Golden Rule

When time order is meaningful,
**plot against time before
ignoring the order.**

Outliers and Data Quality: Always Investigate

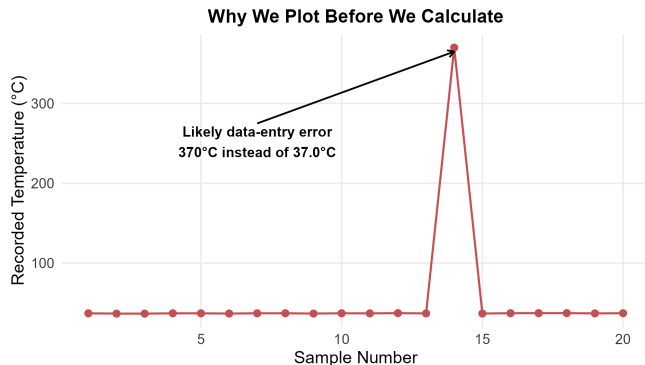
When you see an unusual value, ask:

1. Was the value entered correctly?
2. Are the units correct?
3. Was the instrument working correctly?
4. Is the value scientifically plausible?
5. Could it be a genuine but unusual observation?

Important

An outlier is **not automatically an error**. Do not remove data simply because it looks unusual.

Example 1.7: A Data-Entry Error



Most body temperatures are near 37°C, but one value is recorded as **370°C**.

- The graph makes the error obvious.
- Without plotting first, the typo could badly distort numerical summaries.

Common Data Problems That Graphs Can Reveal

Problem	Example	Possible Visual Signature
Typing error	19.4 entered as 194	One extreme observation
Wrong unit	Some masses in mg instead of g	Two separated groups or a huge gap
Missing-value code	Missing recorded as -99	Strange stack at -99
Instrument drift	Calibration changes over days	Gradual trend in a time plot
Duplicated record	Same row copied twice	Unusually repeated points

Graphs are tools for checking data quality, not only for presenting results.

Apply Your Knowledge 1.1: Breakfast Cereals

Consider a dataset describing 77 breakfast cereals:

Brand	Manufacturer	Type	Calories	Sugar (g)
All Bran	K	C	70	5
All Bran Fiber	K	C	50	0
Almond Delight	R	C	110	8
Apple Cheerios	G	C	110	10
Apple Jacks	K	C	110	14

Questions:

1. What are the individuals?
2. Which variables are categorical? Which are quantitative?
3. Which graph would you use for Manufacturer?
4. Which graph would you use for Sugar across all 77 cereals?

Apply Your Knowledge 1.1: Answers

- **Individuals:** the 77 cereal brands.
- **Categorical:** Manufacturer and Type.
- **Quantitative:** Calories and Sugar.
- **Manufacturer:** bar graph.
- **Sugar for 77 cereals:** histogram is a good starting choice.

Reasoning Pattern

Do not memorize graph names in isolation. Start by identifying the **individual**, then the **variable type**, then the **question**.

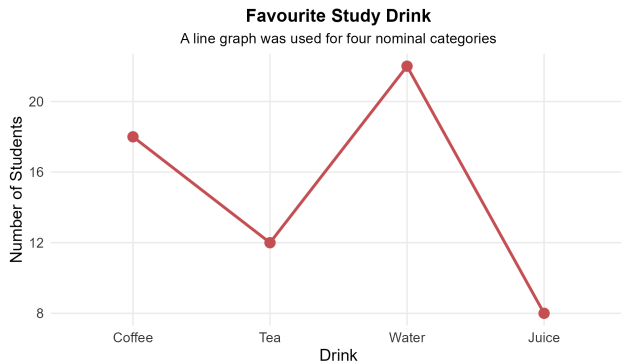
Apply Your Knowledge 1.2: Choose the Graph

Question	Structure	Best Starting Graph
Which blood type is most common?	Categorical	Bar graph
How are four categories divided within one whole?	Categorical, part-to-whole	Pie or bar graph
What is the distribution of 150 plant heights?	Quantitative	Histogram
What are the exact reaction times of 12 participants?	Quantitative, small sample	Dotplot
Is river temperature changing throughout the day?	Quantitative over time	Time plot

Master Guide: Which Graph Should You Use?

Data Situation	Graph	What to Describe	Common Mistake
Categorical counts / percentages	Bar	Largest/smallest categories; comparisons	Connecting categories with a line
Parts of one whole	Pie	Dominant slices; part-to-whole	Using separate group rates as slices
Small quantitative sample	Dotplot	Individual values; SOCS	Hiding a tiny sample in bins
Larger quantitative sample	Histogram	SOCS; overall shape	Poor bin choice
Data recorded over time	Time plot	Trend; cycles; sudden changes	Ignoring time order

Practice 1 (Easy): What Is Wrong with This Graph?



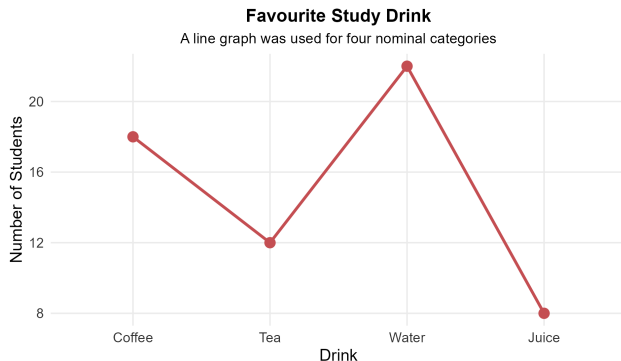
Your Task

The graph shows students' **favourite study drink**. Identify the main problem with the display.

Ask yourself:

- What type of variable is *drink*?
- Does connecting the categories imply something that is not real?

Practice 1 (Easy): What Is Wrong with This Graph?



Your Task

The graph shows students' **favourite study drink**. Identify the main problem with the display.

Ask yourself:

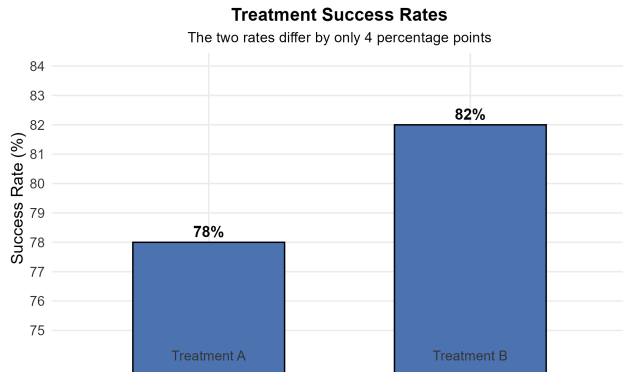
- What type of variable is *drink*?
- Does connecting the categories imply something that is not real?

Answer

Drink is nominal categorical.

Connecting the categories suggests an order or continuous movement that does not exist. Use a **bar graph**.

Practice 2 (Medium): Is the Difference Really That Large?



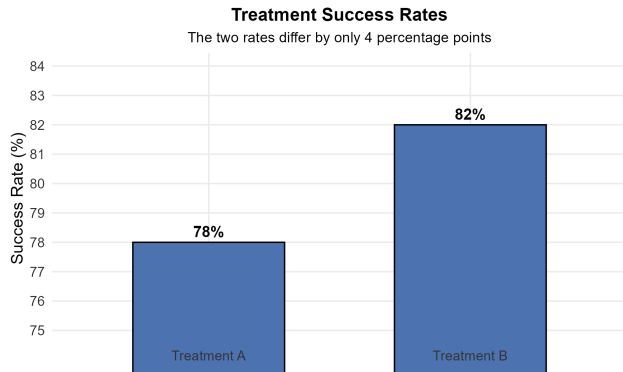
Your Task

Treatment A has a success rate of **78%**; Treatment B has a success rate of **82%**. What makes this graph misleading?

Look carefully at:

- the vertical axis
- the visual size of the difference
- the actual numerical difference

Practice 2 (Medium): Is the Difference Really That Large?



Your Task

Treatment A has a success rate of **78%**; Treatment B has a success rate of **82%**. What makes this graph misleading?

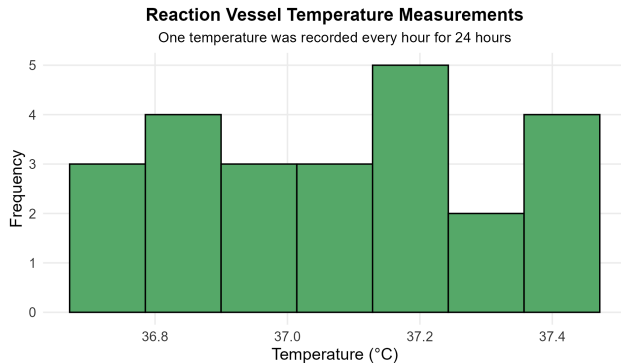
Look carefully at:

- the vertical axis
- the visual size of the difference
- the actual numerical difference

Answer

The axis begins near **75%**, not 0, which exaggerates the **4-point difference**. Bar charts should generally start at zero.

Practice 3 (Hard): What Important Information Is Missing?



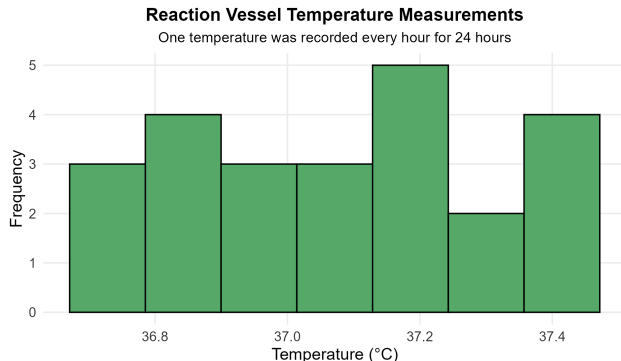
Research Question

A temperature was recorded every hour for 24 hours. The researcher wants to know:

Did the reaction vessel temperature drift during the experiment?

What important information has the graph discarded?

Practice 3 (Hard): What Important Information Is Missing?



Research Question

A temperature was recorded every hour for 24 hours. The researcher wants to know:

Did the reaction vessel temperature drift during the experiment?

What important information has the graph discarded?

Answer

The histogram shows the temperature distribution but removes **time order**. It cannot reveal drift, cycles, or sudden changes. Plot **temperature versus time** first.

Looking Ahead: Chapter 2

In Chapter 1, we learned to **see** center and spread. In Chapter 2, we will learn to **calculate** them.

Mean Median

and measures of variability such as

Range Interquartile Range Standard Deviation.

The Same Principle Continues

Numbers and graphs should support each other.